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**Speech-to-Text Sentiment Analysis: Analyzing
Emotions in Spoken Language**

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1. Project Overview

This project aims to develop an integrated system that not only transcribes spoken language but also performs deep, context-aware sentiment and emotion analysis. Moving beyond basic positive/negative classification, we will analyze discrete emotions directly from audio input. The final deliverable will be a functional pipeline that accepts audio and outputs a rich sentiment profile, visualizing both lexical content and paralinguistic cues.

2. Objectives

Primary: To build a robust Speech-to-Text (STT) module using a state-of-the-art transformer-based model for accurate transcription.

Core Analysis: To implement a multi-class sentiment and emotion classifier that operates on the transcribed text, leveraging pre-trained Large Language Models (LLMs) fine-tuned for emotional intelligence. To explore and integrate prosodic features (pitch, tone, speech rate) from the audio to augment text-based analysis, improving detection of emotions.

Integration & Delivery: To create a seamless end-to-end speech-to-text sentiment analysis pipeline with a simple user interface.

3. Methods and Technologies

The following are methods and technologies we are going to apply throughout the project:

Speech-to-Text: Utilize OpenAI's Whisper model (specifically the small or base variant for efficiency) for its high accuracy and robustness to background noise.

Textual Sentiment Analysis: Fine-tune RoBERTa, a pre-trained transformer model, on a combined dataset for multi-label emotion classification.

Audio Feature Augmentation: Use audio libraries to extract fundamental frequency (F0), energy, and speech rate. We will use methods to fuse these features with the textual embeddings to create a multi-modal model.

Integration & Evaluation: Integrate modules using a Python-based pipeline. Evaluate STT with Word Error Rate (WER) and sentiment model performance with F1-score (macro-average due to class imbalance) and accuracy.

4. Expected Outcomes

By the project deadline, we will deliver:

- A trained sentiment/emotion classification model (weights and tokenizer) capable of identifying core emotions.
- Pure text-based sentiment and audio-augmented analysis.
- A simple demonstration interface where a user can enter audio and receive a synchronized transcript annotated with sentiment and emotion tags.